

Week 5

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Recap

Compare the means of multiple treatment groups to determine if our treatment factor (manipulation) had an effect on the response variable (outcome). To do this, we use the following statistical model:

$$Y_{ij} = \mu + A_i + e_{ij}$$

where

$i = 1, \dots, a$ (a = number of treatments)

$j = 1, \dots, r$ (r = number of replicates)

Y_{ij} = response value of the j^{th} unit receiving treatment i

μ = overall or general mean

A_i = effect of the i^{th} level of treatment factor A

e_{ij} = random error with $e_{ij} \sim N(0, \sigma^2)$

This model is equivalent to a regression model

- This model is a regression model with a single categorical variable but just parameterised differently.
- We use μ and A_i 's
- Regression uses β_0 and β_i 's
 - β_0 is the mean of the baseline category
 - β_i is the difference between the mean of category i and the baseline category
- In both cases the data are the same
 - a groups and
 - we are interested in the mean response in these groups
 - and differences between the groups

Why ANOVA model?

- When all explanatory variables are categorical - ANOVA is convenient
- The A_i notation is more concise
- Leads directly to the Analysis of Variance

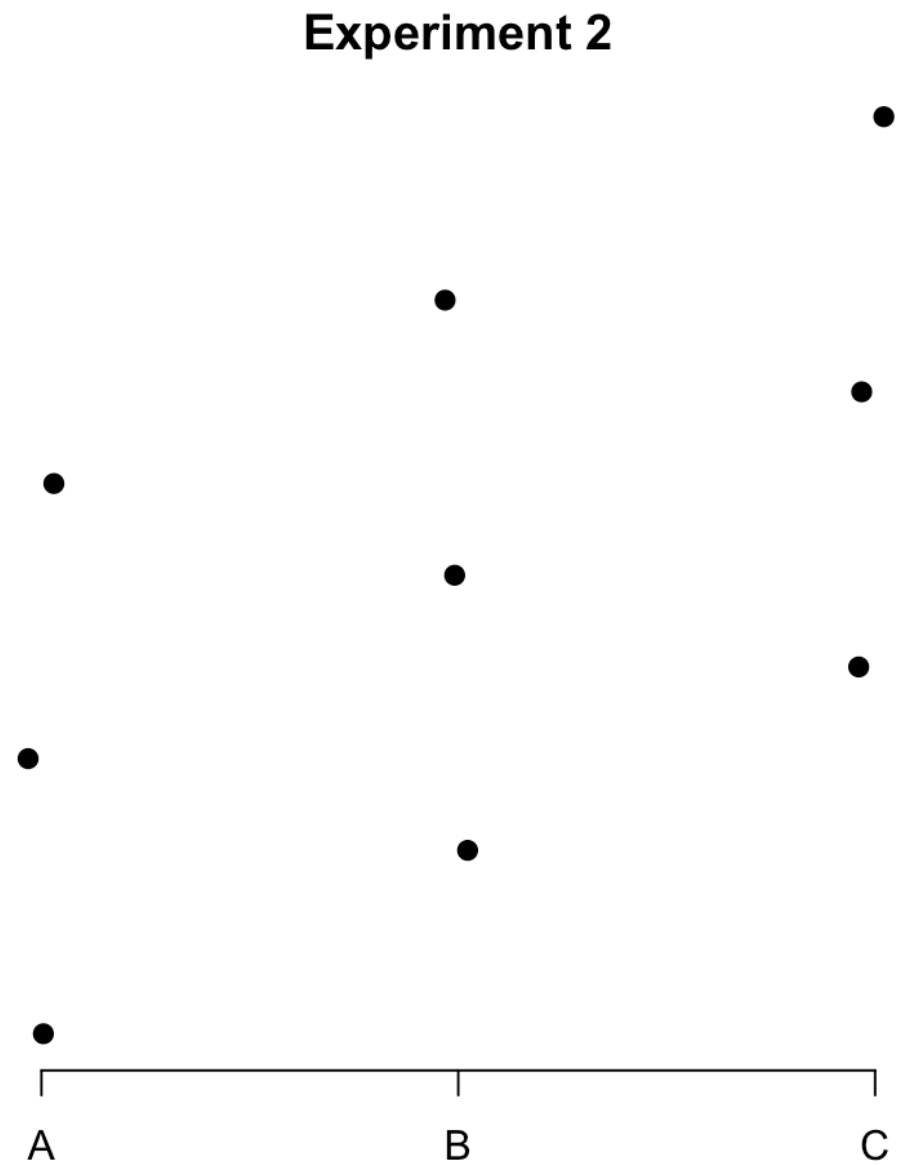
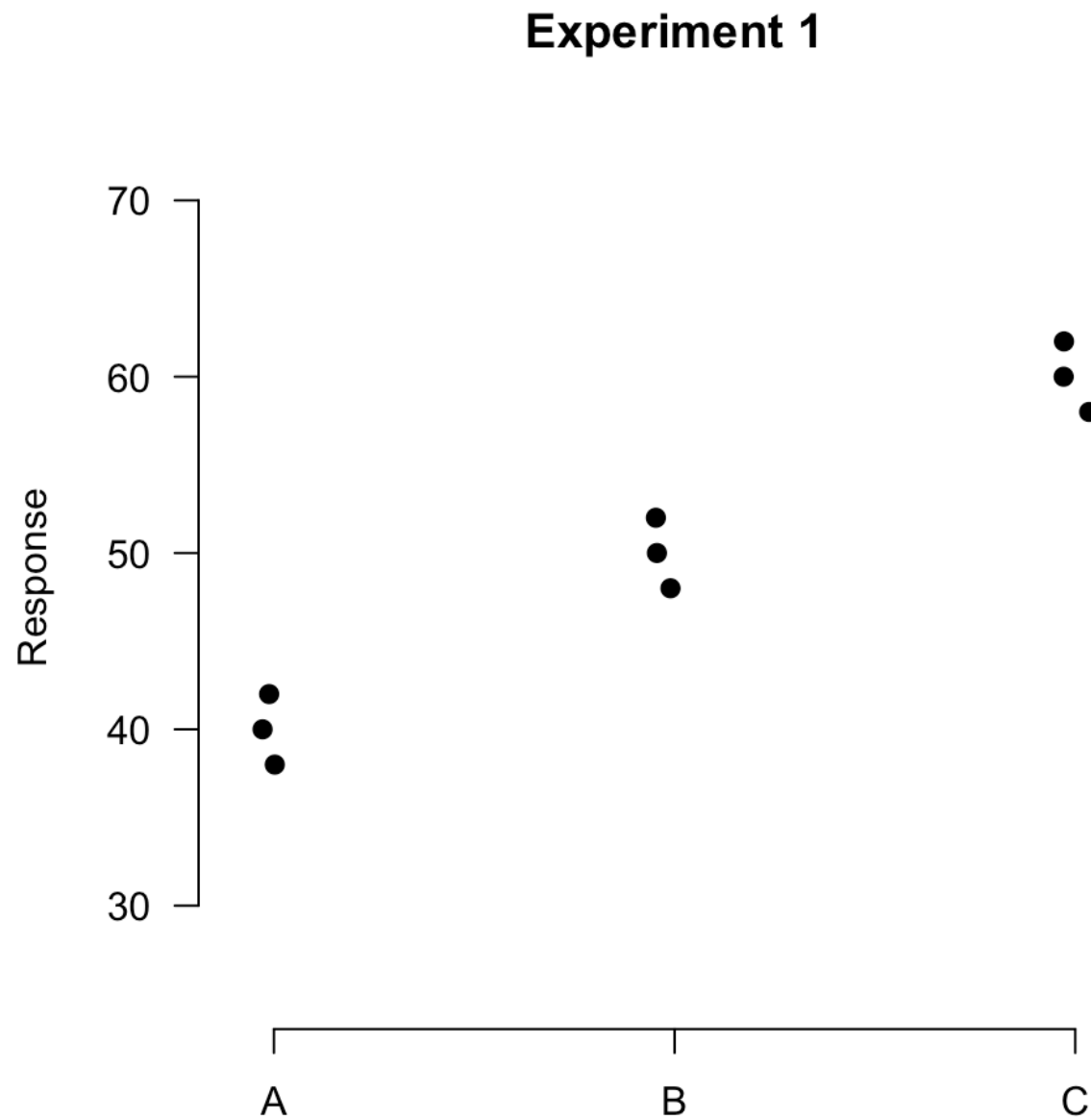
ANOVA

Table 1: Hypothetical Experiment

<i>(a) Experiment 1</i>			
Treatment	A	B	C
	40	48	58
	42	50	62
	38	52	60
Average	40	50	60

<i>(b) Experiment 2</i>			
Treatment	A	B	C
	40	65	45
	25	35	75
	55	50	60
Average	40	50	60

Plot of hypothetical data



Interpretation

- Both experiments have the same means
- But the variability among observations within treatments is much larger in experiment 2 than 1
- So we are more likely to conclude that the treatment means differ in experiment 1 than in experiment 2 due to this difference
- This is the basis of the ANOVA test

ANOVA

ANOVA compare the between-treatment (signal) to the within-treatment (noise) variation as a ratio of the two:

$$\frac{\text{Between variation (differences in group means)}}{\text{Within variation (differences within groups)}}$$

- **Large ratios** imply signal (difference among the means) is **large** relative to noise (variation within groups)
→ evidence for a difference in the means.
- **Small ratios (close to 1)** imply signal is **small** relative to noise
→ no evidence that means differ.

How do we measure the between and within variation?

- There are many ways in which we can measure variability (e.g. IQR, standard deviation, range, etc.)
- One common way is the **sums of squares** (SS) which is the sum of squared deviations from a mean.
- The numerator of the formula for variance is the sums of squares of observations around the overall mean:

$$\text{Variance} = \frac{\text{Sums of Squares}}{\text{Degrees of Freedom}} = \frac{\sum (Y_{ij} - \bar{Y})^2}{N - 1}$$

Sums of squares

Using our model we can decompose the total sums of squares $\sum (Y_{ij} - \bar{Y})^2$ into:

- Sums of squares between groups / due to treatments (SSB, SS_{TREAT}, SSA)
- Sums of squares within groups (SSW/SSE)

Decomposition of the Total Sum of Squares

$$Y_{ij} = \mu + A_i + e_{ij}$$

- Rewrite in terms of means. Replace A_i with $(\mu_i - \mu)$ and e_{ij} with $(Y_{ij} - \mu_i)$

$$Y_{ij} = \mu + (\mu_i - \mu) + (Y_{ij} - \mu_i)$$

- Use estimates

$$Y_{ij} = \bar{Y} + (\bar{Y}_{i.} - \bar{Y}) + (Y_{ij} - \bar{Y}_{i.})$$

- Move overall mean to LHS

$$Y_{ij} - \bar{Y} = (\bar{Y}_{i.} - \bar{Y}) + (Y_{ij} - \bar{Y}_{i.})$$

- Square and sum across indices

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i \sum_j (\bar{Y}_{i.} - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_{i.})^2$$

- Simplify

$$\sum_i \sum_j (Y_{ij} - \bar{Y})^2 = \sum_i r (\bar{Y}_i - \bar{Y})^2 + \sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$$

$$SS_{\text{Total}} = SS_{\text{Between Groups}} + SS_{\text{Within Groups}}$$

Apply to hypothetical example (Experiment 1)

From sums of squares to variance

- When we divide sums of squares by the appropriate degrees of freedom, we get mean squares.

$$MS = \frac{SS}{df}$$

- This turns a sum of squared deviations into an average squared deviation, which is a measure of variance.

From sums of squares to variance

- Mean squares for group differences:

$$MSB = \frac{SSB}{a - 1}$$

- Mean squares for within group differences:

$$MSE = \frac{SSE}{N - a}$$

The F-ratio

Then when we take the ratio of two mean squares, we get the F-ratio:

$$F = \frac{MSB}{MSE}$$

F-ratio and our hypothesis

- When $MSB \approx MSE$ then there is no evidence against

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4 = \mu$$

- When $MSB \gg MSE$ then there is evidence against H_0 suggesting

$$H_1 : \text{at least one of the treatment means } (\mu_i) \text{ differ}$$

Apply to hypothetical example (Experiment 1)

But how large is large enough?

Need to perform a hypothesis test:

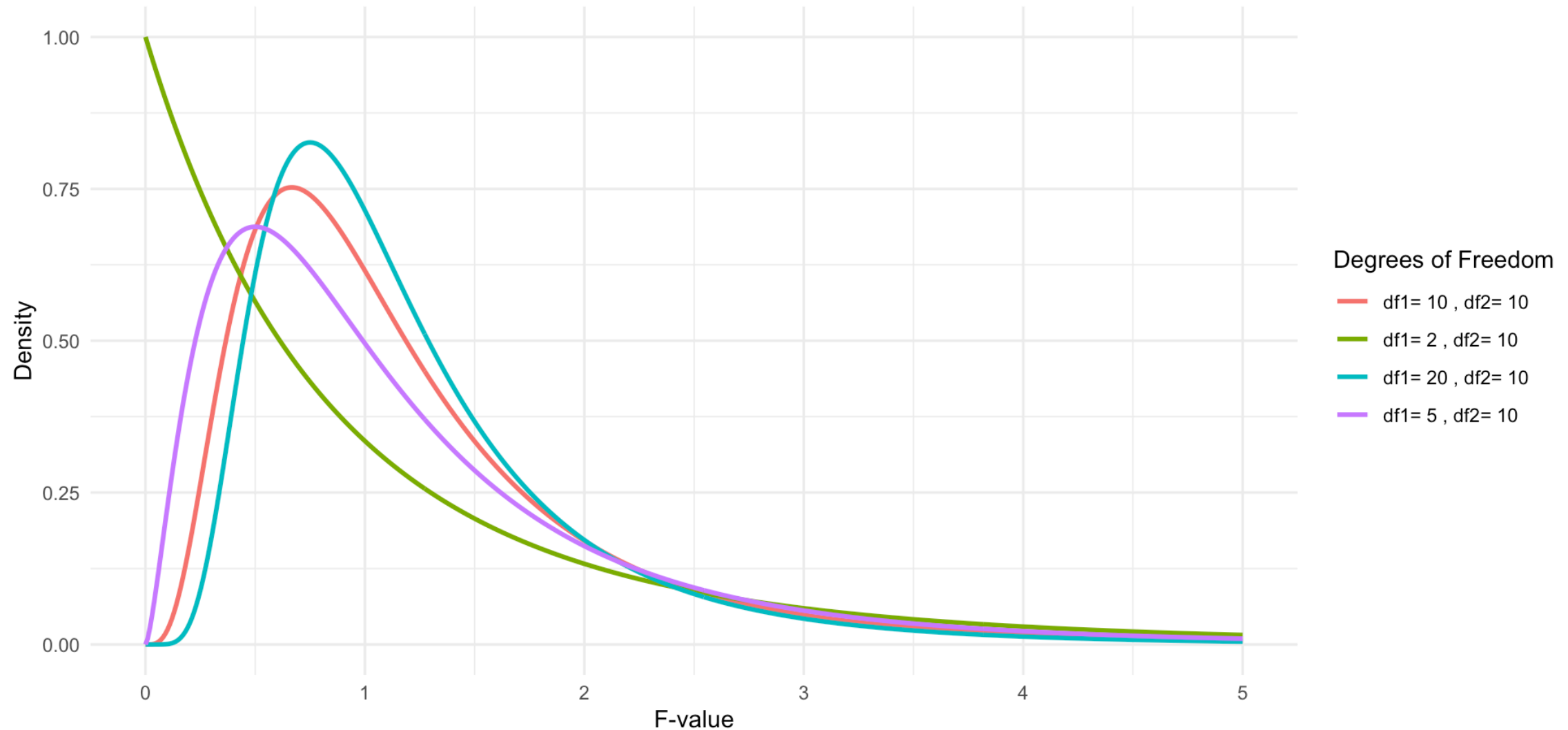
- Get sampling distribution of test statistic
- Compare test statistic to sampling distribution
- Get p-value (using R or tables)

The F-ratio follows an F-distribution under the null hypothesis of equal means:

$$\frac{MSB}{MSE} \sim F_{df1, df2} \sim F_{a-1, N-a}$$

Meet the F-distribution

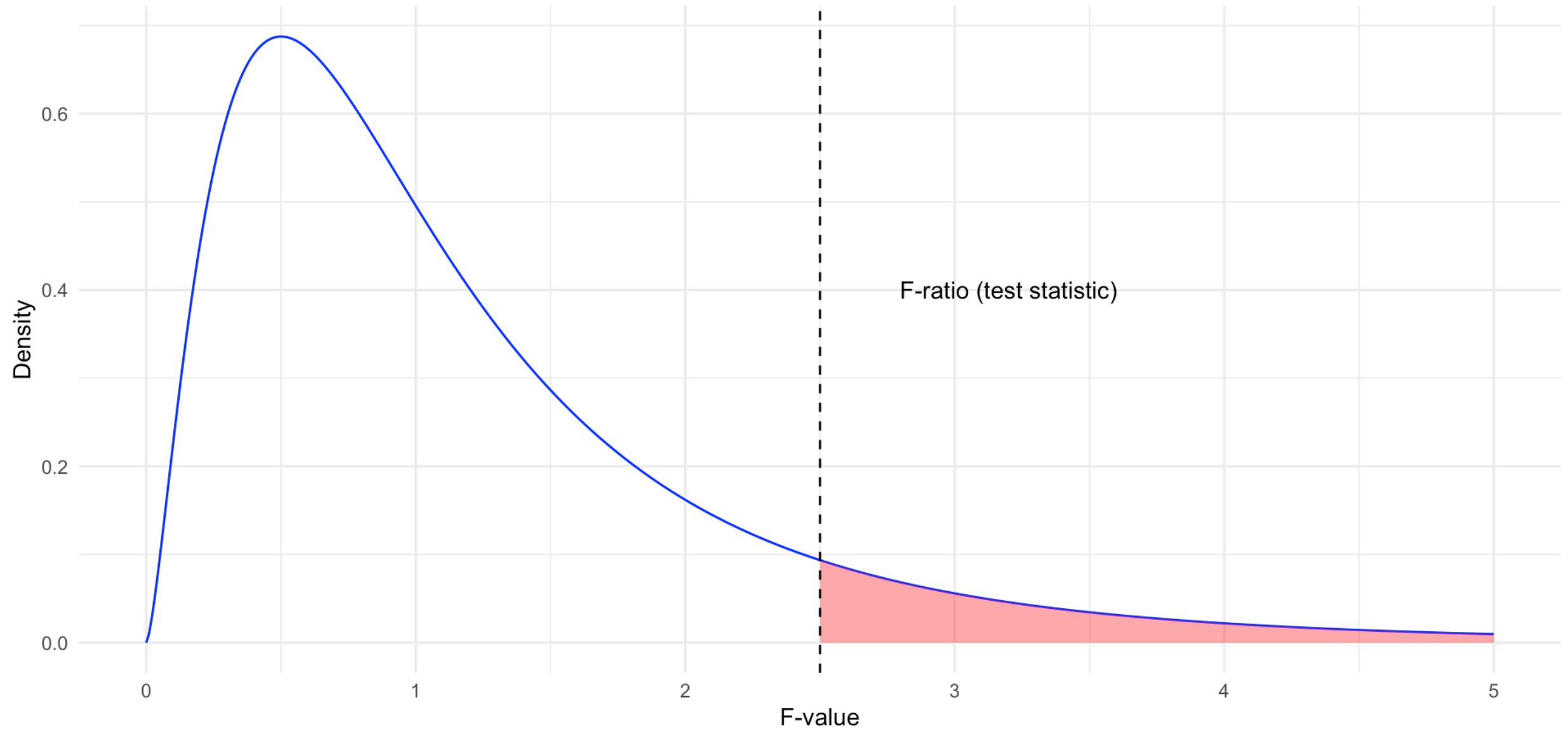
Shape depends on the degrees of freedom of the numerator and denominator



Get the p-value and make a conclusion

- P-value: Probability of observing our test statistic (or more extreme) under the null hypothesis.
- Small p-value = it is unlikely to observe this test statistic if the null hypothesis was true.
- Large p-value = it is likely to observe this test statistic if the null hypothesis was true.

Get the p-value and make a conclusion



P-value in R

```
1 pf(q = 75, df1 = 2, df2 = 6, lower.tail = FALSE)
```

```
[1] 5.689577e-05
```

Summarise the hypothesis test in an ANOVA table

Source	Sums of Squares (SS)	df	Means Squares (MS)	F
Treatment	$\sum_i r(\bar{Y}_i - \bar{Y})^2$	$a - 1$	$MSB = \frac{SS_A}{(a-1)}$	$\frac{MSB}{MSE}$
Residuals (Error)	$\sum_i \sum_j (Y_{ij} - \bar{Y}_i)^2$	$N - a$	$MSE = \frac{SSE}{(N-a)}$	
Total	$\sum_i \sum_j (Y_{ij} - \bar{Y})^2$	$N - 1$		

Hypothetical example ANOVA table (Experiment 1)

Source	Sums of Squares (SS)	df	Means Squares (MS)	F
Treatment	600	2	300	75
Residuals (Error)	24	6	4	
Total	624	9		

Playback Experiment in R

```
1 summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Condition	3	634	211.2	1.3	0.276
Residuals	176	28587	162.4		

Conclusion

The ANOVA results indicate there is no significant effect of the playback condition on mean comprehension score ($F = 1.3$, $p\text{-value} = 0.276$). This suggests that playback speed does not impact comprehension of audio-only content in this experiment.

Another Example

A researcher in music therapy is investigating how different types of background music influence stress reduction in young adults. Participants are randomly assigned to listen to one of three types of music for 30 minutes after completing a mentally demanding task:

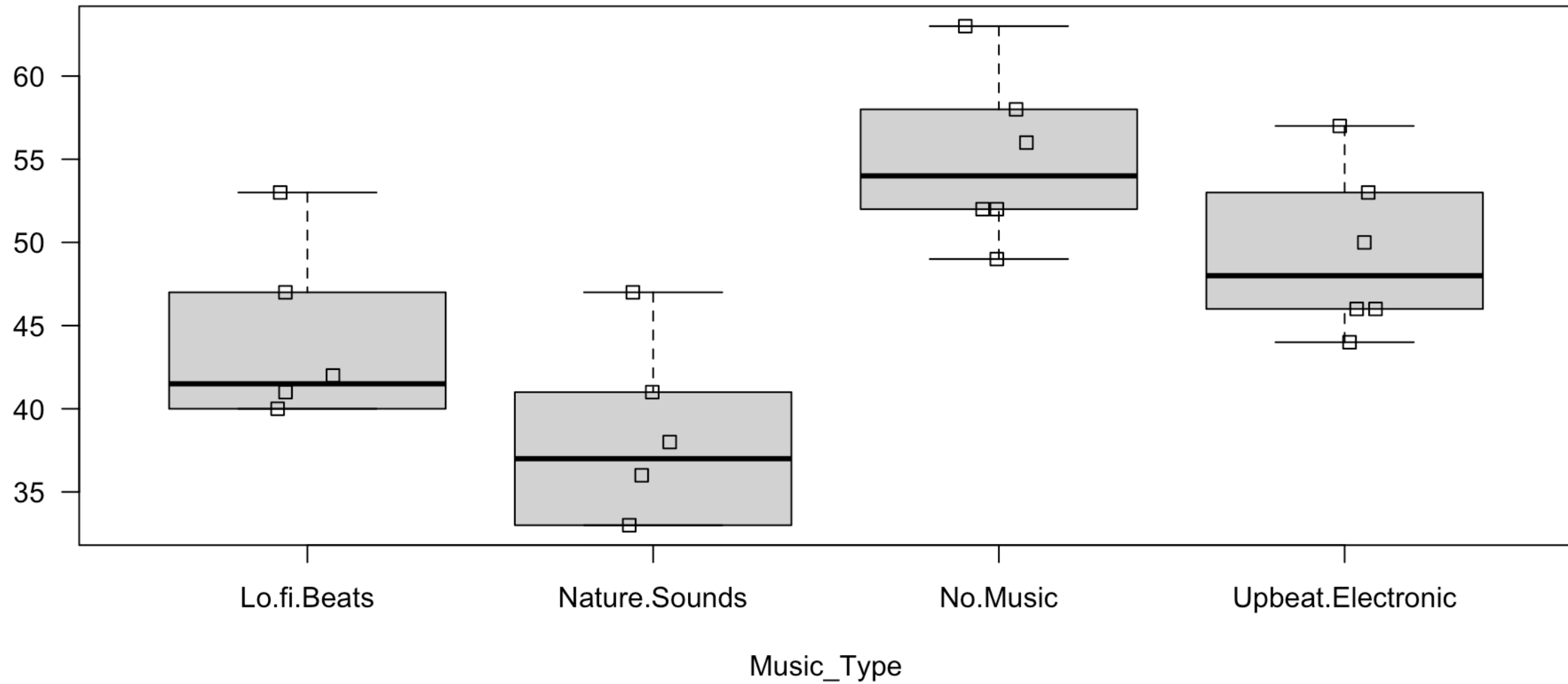
1. Lo-fi beats – smooth, downtempo music commonly used for relaxation and study
2. Nature sounds with soft instrumental music – a combination of birdsong, rain, and calming melodies
3. Upbeat instrumental electronic music – energetic but non-distracting music
4. Control - no music.

After listening, participants complete a stress level assessment using a standardized questionnaire that scores their stress levels from 0 (completely relaxed) to 100 (highly stressed).

Data

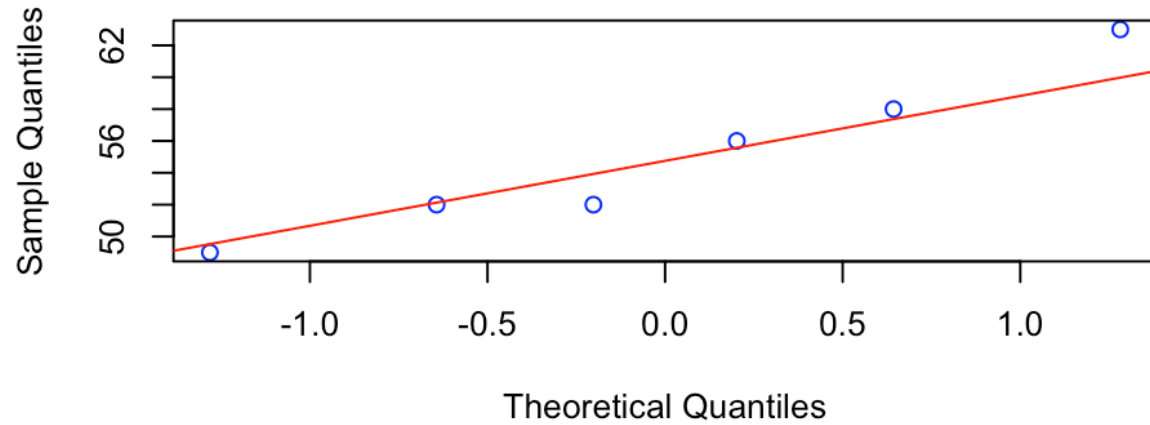
Lo.fi.Beats	Nature.Sounds	No.Music	Upbeat.Electronic
40	36	56	44
53	19	63	46
19	33	49	57
47	38	52	53
41	47	52	46
42	41	58	50

Assumptions

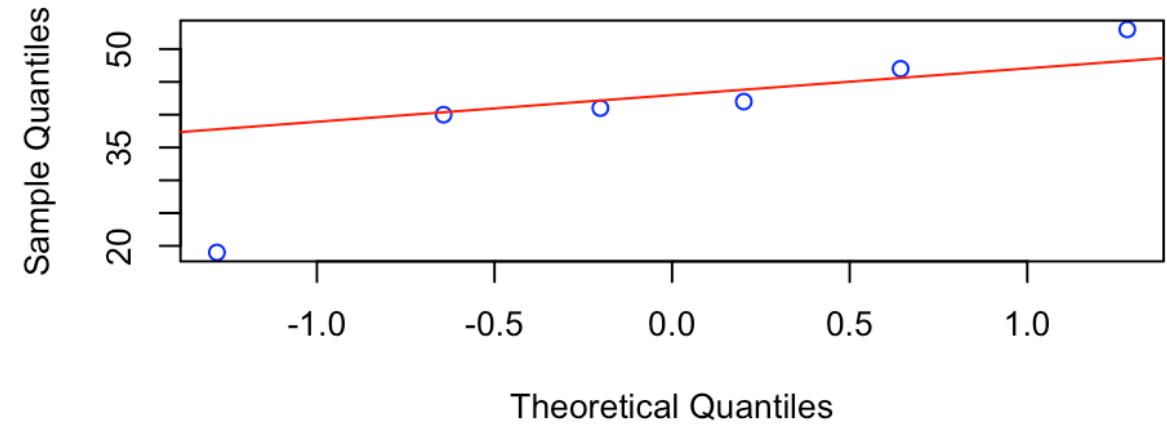


Q-Q Plots

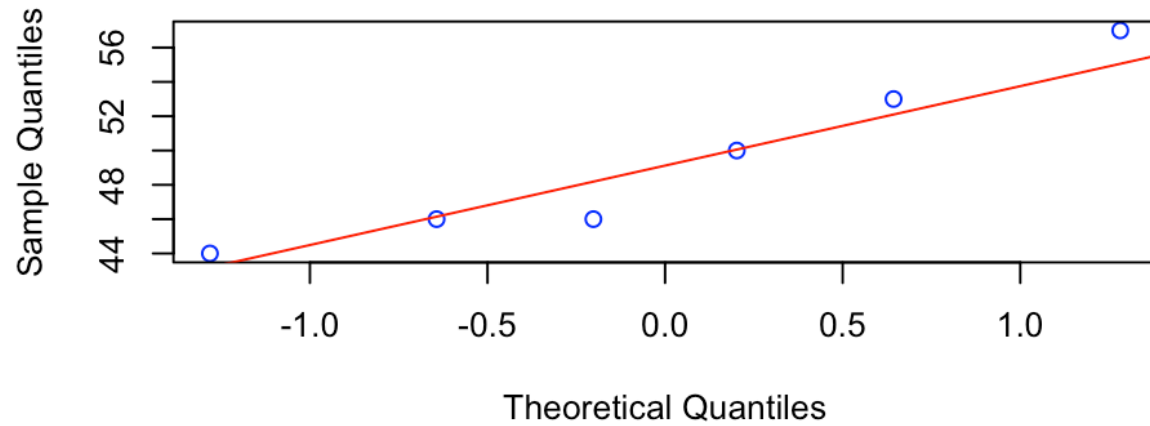
Normal Q-Q Plot



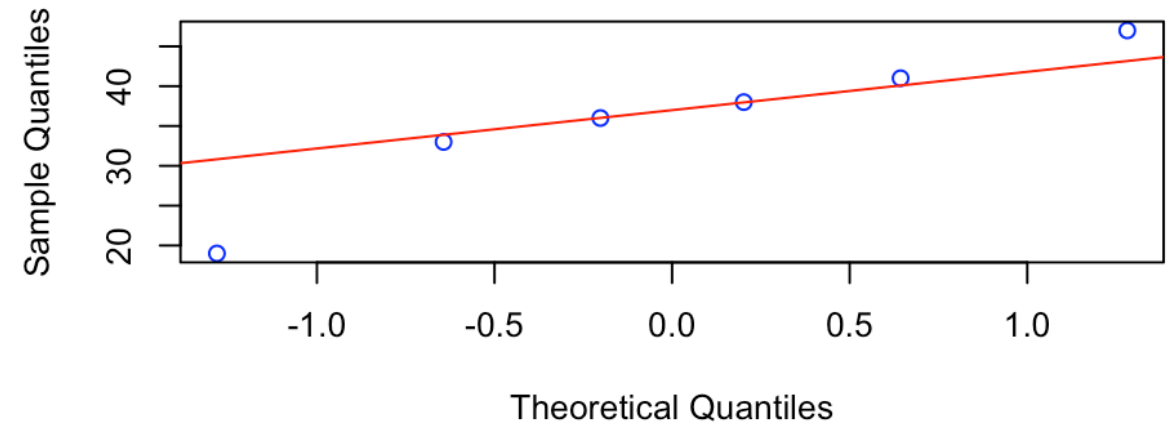
Normal Q-Q Plot



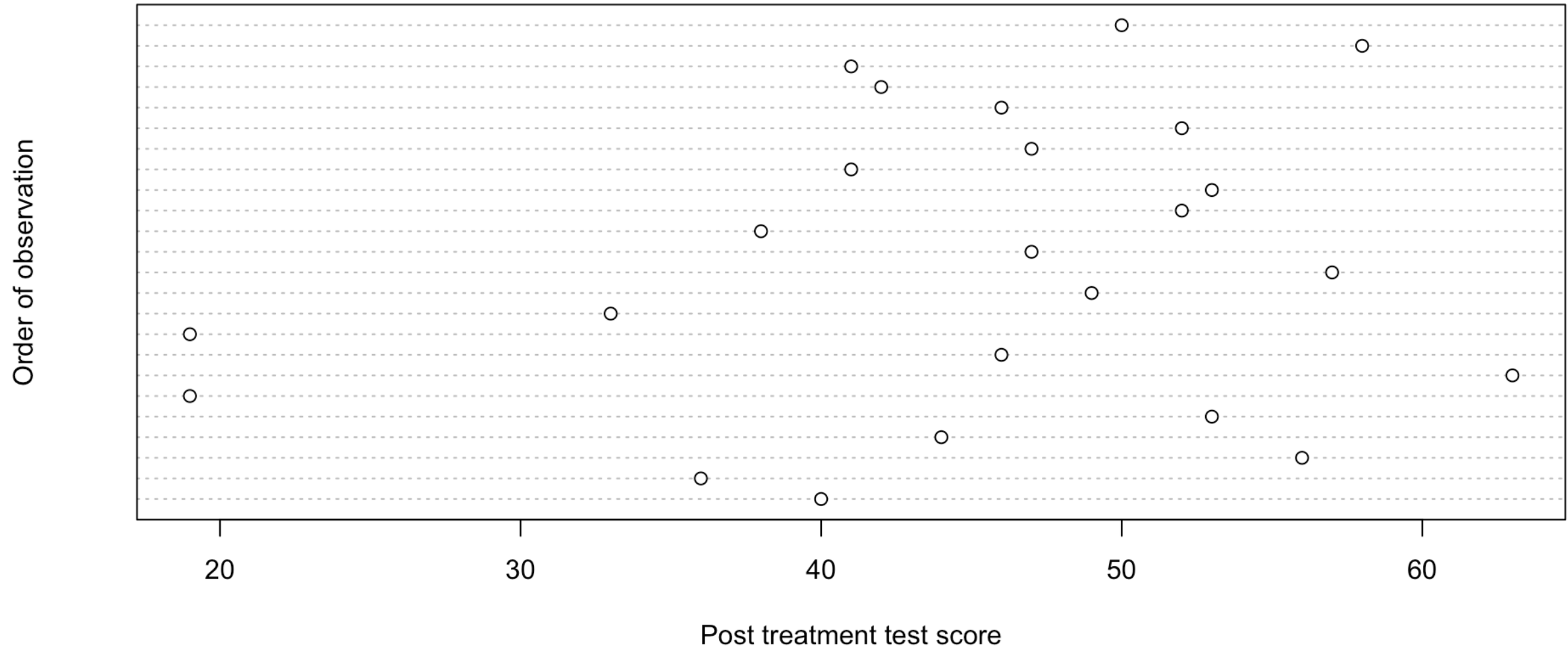
Normal Q-Q Plot



Normal Q-Q Plot



Independence



In R: Exploratory Data Analysis

```
1 head(data_long)
```

```
# A tibble: 6 × 2
```

	Music_Type	Stress_Score
	<fct>	<dbl>
1	Lo.fi.Beats	40
2	Nature.Sounds	36
3	No.Music	56
4	Upbeat.Electronic	44
5	Lo.fi.Beats	53
6	Nature.Sounds	19

In R: Exploratory Data Analysis

```
1 summary(data_long)
```

	Music_Type	Stress_Score
Lo.fi.Beats	:6	Min. :19.00
Nature.Sounds	:6	1st Qu.:40.75
No.Music	:6	Median :46.50
Upbeat.Electronic	:6	Mean :45.08
		3rd Qu.:52.25
		Max. :63.00

```
1 class(data_long$Music_Type)
```

```
[1] "factor"
```

In R: Model Fitting

```
1 model <- aov(Stress_Score ~ Music_Type, data = data_long)
```

Parameter estimates: means

```
1 model.tables(model, type = "means", se = TRUE)
```

Tables of means

Grand mean

45.08333

Music_Type

Music_Type

Lo.fi.Beats

40.33

Nature.Sounds

35.67

No.Music

55.00

Upbeat.Electronic

49.33

Standard errors for differences of means

Music_Type

4.764

replic.

6

Parameter estimates: effects

```
1 model.tables(model, type = "effects", se = TRUE)
```

Tables of effects

Music_Type				
Music_Type	Lo.fi.Beats	Nature.Sounds	No.Music	Upbeat.Electronic
	-4.750	-9.417	9.917	4.250

Standard errors of effects

	Music_Type
	3.369
replic.	6

ANOVA

```
1 summary(model)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Music_Type	3	1366	455.3	6.685	0.00263 **
Residuals	20	1362	68.1		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1